4.7 Distribution System of Water

Distribution of water is the supply of safe and wholesome water to all parts of the area served at adequate pressure and quantity. Hence, the distribution system may consist of the following

1. Pipe lines of different sizes to convey water.
2. Valves for controlling the flow in the pipe lines.
3. Meters for measuring the consumption.
4. Hydrants for providing connections with water mains for releasing water during fires.
5. Service connections to the individual houses.
6. Pumps for lifting and forcing the water into the distribution pipes.
7. Service reservoirs for storing the treated water and feeding to the distribution pipes.

Requirements of Good Distribution system

A good distribution system should satisfy the following general requirements

1. It should be capable of supplying water in adequate quantities and pressure at all points of the area served.
2. It should meet the demands of water supply for fire-fighting purposes.
3. It should be thoroughly reliable.
4. It should be economical in its design, layout and construction.
5. It should be easy and simple to operate and repair.
6. It should be safe against any future pollution of water.
7. It should be water-tight so as to keep the "losses due to leakage" to the minimum.
8. It should be safe and not cause the failure of the pipelines by bursting, etc.

Different Systems of supplying water

Depending upon the situation, the water supply system may be classified as below

1. Methods of distribution
2. Systems of water supply
3. Methods of Layout of distribution pipes

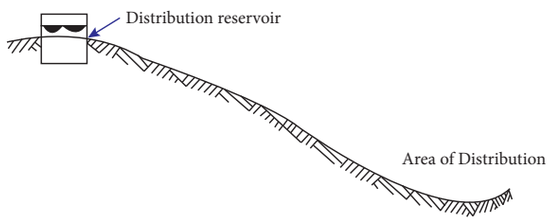
4.7.1 Methods of distribution

Depending upon the levels of water source and that of the area served, topography of the area served and other local conditions, following methods are adopted.

a. Gravity system
b. Pumping system
c. Combined pumping and gravity system

a. Gravity system

In this system as shown in fig, treated water is conveyed at atmospheric pressure through pipes by gravity. This system is adopted when the source of water supply is situated at a higher level than the area served.



Merits

i. Most safe, reliable and economical.
ii. No pumping is required.

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